

Enzyme Engineering & Directed Evolution Planner: An Agentic AI System for Rational Protein Design

Problem Statement & Motivation

Enzyme engineering projects aimed at improving thermostability, catalytic activity, solvent tolerance, or substrate specificity represent a critical bottleneck in both academic research and industrial biotechnology. Despite advances in computational tools and high-throughput screening technologies, the early design phase of directed evolution campaigns remains inefficient and error-prone. Researchers frequently make suboptimal decisions about mutation targets, library construction strategies, and screening approaches because they lack a unified framework for synthesizing structural data, evolutionary conservation patterns, and experimental constraints.

The core business problem is straightforward: enzyme engineering campaigns often fail or require excessive iteration cycles because initial design decisions are made without comprehensive analysis of available evidence. Common failure modes include selecting structurally inappropriate mutation sites, designing combinatorial libraries that exceed realistic screening capacity, underestimating the complexity of library construction, and failing to document the rationale behind design choices in ways that enable productive iteration. These inefficiencies translate directly into wasted reagents, lost researcher time, and delayed project timelines—costs that are particularly burdensome for academic labs with limited budgets and early-stage biotech companies operating under tight resource constraints.

The target users for this system are undergraduate and graduate researchers conducting enzyme engineering projects, academic research groups running directed evolution campaigns, teaching laboratories implementing protein engineering modules, and small biotech R&D teams performing rapid enzyme optimization for industrial applications. These users share a common need: they require structured, evidence-based guidance that respects their experimental constraints while remaining flexible enough to accommodate diverse objectives and available resources.

An agentic AI solution is uniquely appropriate for this problem domain because enzyme engineering design is inherently a multi-step, tool-using process that benefits from specialized reasoning at each stage. The workflow begins with goal interpretation, proceeds through sequence and structure analysis, generates candidate mutation strategies, estimates experimental feasibility, proposes screening approaches, and concludes with critical review of internal consistency. This process is dynamic rather than static—optimal choices depend heavily on user-specific constraints such as screening throughput, assay type, expression host, and available instrumentation. A traditional chatbot or single-pass language model cannot effectively handle this complexity because it cannot delegate specialized subtasks, maintain structured intermediate representations, and perform iterative refinement through multi-agent critique. The agentic approach enables the system to decompose the problem into manageable components, apply domain-specific reasoning at each step, and produce outputs that are both evidence-grounded and internally consistent.

Use Cases & Scenarios

To illustrate the practical utility of this system, I present three concrete user scenarios that span common enzyme engineering objectives and experimental contexts. In the first scenario, a graduate student aims to

improve the thermostability of a model enzyme by approximately ten degrees Celsius but faces a significant constraint: their laboratory can screen only about two thousand variants per week using a plate-based colorimetric assay. The student provides the enzyme sequence and either an experimental crystal structure or an AlphaFold-predicted model. The agentic system parses this objective and constraint set, then orchestrates a multi-agent workflow. The Structure Reasoning Agent identifies candidate mutation regions by analyzing flexible loops, helix caps, and core packing interactions that are known to influence protein stability. The Sequence Annotation Agent computes conservation scores from homologous sequences to flag positions where mutations are likely to be tolerated. The Library Design Agent then proposes two or three feasible library construction strategies—for example, a focused site-saturation library targeting six to twelve key residues, or a smaller combinatorial library exploring pairwise interactions among four high-confidence positions. Each library option is accompanied by an estimated variant count and projected screening burden, allowing the user to make an informed decision that balances exploration breadth against practical capacity. The system produces a ranked design slate with explicit rationale for each recommendation and maintains an assumption log that documents uncertainties and data gaps.

The second scenario involves substrate specificity engineering. A postdoctoral researcher wants to shift an enzyme's preference from its native substrate to an alternative compound, and preliminary literature suggests that several active-site pocket residues are likely determinants of specificity. The researcher uploads the enzyme sequence, structure, and a brief description of the desired specificity change. The Intake Agent normalizes this goal into measurable criteria, such as improving the ratio of catalytic efficiency for the target substrate relative to the native substrate. The Sequence Annotation Agent retrieves known active-site residues from database annotations and literature sources, while the Structure Reasoning Agent maps these residues onto the three-dimensional model and identifies second-shell positions that may influence pocket geometry. The Library Design Agent proposes a two-round strategy: an initial broad scan of pocket residues using site-saturation mutagenesis to identify beneficial single mutations, followed by a focused combinatorial library that explores synergistic interactions among the top hits from round one. The Assay Strategy Agent suggests evaluation criteria and control experiments, such as measuring activity against both substrates to quantify specificity shifts and including wild-type enzyme as a reference point. Critically, the Critic Agent reviews the entire plan and flags potential issues, for example, if the proposed library size exceeds stated throughput constraints, or if the assay lacks sufficient dynamic range to detect subtle specificity changes. This multi-agent critique loop ensures that the final plan is both scientifically sound and experimentally feasible.

The third scenario demonstrates the system's capacity for iterative design. After completing an initial round of mutagenesis and screening, a researcher has activity measurements for ninety-six enzyme variants and has identified several improved mutants. The researcher uploads these results as a structured CSV file and asks the system to recommend a second-round design. The system ingests the data and performs pattern analysis to identify which positions contributed most strongly to improved activity, whether there are hints of epistatic interactions among mutations, and whether activity gains came at the cost of reduced stability or expression levels. Based on this analysis, the Library Design Agent proposes a refined strategy for round two—either a combinatorial library that explores interactions among winning positions, or a stability rescue library if the data suggest that beneficial activity mutations have destabilized the protein. The system generates an updated campaign plan with revised feasibility estimates and evaluation metrics tailored to the second round objectives. This iterative capability transforms the

system from a one-time planning tool into a persistent design partner that learns from experimental outcomes and adapts recommendations accordingly.

Initial System Design

The system architecture centers on a user-facing web application that orchestrates a multi-agent workflow to produce structured campaign plans. These plans include recommended mutation targets, library construction strategies, feasibility estimates, and evidence-backed rationale, all organized into exportable artifacts that support documentation and reproducibility. The multi-agent design is essential for meeting the Milestone II requirement and reflects the inherently modular nature of enzyme engineering decision-making.

The Intake and Constraint Agent serves as the system's entry point, collecting minimal required inputs such as the engineering objective, sequence and structure availability, screening throughput capacity, assay type, and expression host constraints. This agent normalizes user-stated goals into measurable targets. For example, translating "make it more stable" into specific metrics like increased melting temperature or improved residual activity after heat challenge. The Sequence and Annotation Agent retrieves or parses protein metadata, including domain boundaries, catalytic residues, post-translational modification sites, and signal peptides. When homologous sequences are available, this agent computes conservation scores that help distinguish positions where mutations are likely to be tolerated from those where changes may disrupt essential function. The Structure Reasoning Agent maps candidate residues onto three-dimensional context, distinguishing core positions from surface-exposed sites, identifying residues proximal to the active site or substrate-binding pocket, and flagging flexible loops or hinge regions that may be amenable to stabilization through rigidification strategies.

The Library Design Agent translates these structural and evolutionary insights into practical experimental plans. It proposes specific library construction options, such as site-saturation mutagenesis at selected positions, focused combinatorial libraries exploring pairwise or higher-order interactions, or broader random mutagenesis approaches when structural information is limited. Crucially, this agent estimates library sizes and screening burden for each option, enabling users to select strategies that match their experimental capacity. The Assay and Screening Strategy Agent complements this by suggesting appropriate screening versus selection frameworks, defining control experiments, and specifying what data should be captured to enable productive iteration in subsequent rounds.

The Critic and Quality Assurance Agent play a central role in the multi-agent workflow required for Milestone II. This agent verifies internal consistency across all recommendations, checking whether proposed library sizes respect stated throughput constraints, whether the chosen assay is well-matched to the engineering objective, and whether key assumptions have been left implicit rather than documented. The Critic produces a risk register that highlights uncertainties and requests clarifications when confidence is low, effectively serving as an automated peer reviewer that catches logical inconsistencies and experimental design flaws before they propagate into the laboratory. Finally, the Report Generator Agent compiles all outputs into a structured document that includes ranked mutation zones, library options with feasibility metrics, an explicit assumption log, citations to supporting literature and database sources, and exportable tables in CSV or JSON format for downstream use.

The multi-agent orchestration follows a planner-specialist-critic-finalizer pattern. The Intake Agent decomposes the user's request into subtasks and assigns them to specialist agents. These specialists return structured outputs such as ranked residue lists, library construction options, and feasibility estimates. The Critic Agent reviews these outputs for consistency and triggers a revision loop if significant issues are detected. Once the critique is satisfied, the Report Generator produces the final campaign plan artifact. This architecture ensures that the system leverages specialized reasoning at each stage while maintaining global coherence through explicit critique and revision mechanisms.

Expected Tools, APIs, and Data Sources

The system relies on a combination of public biological databases, computational utilities, and user-provided inputs. Core data sources include protein sequence and annotation databases such as UniProt, domain databases like Pfam and InterPro, and structural resources including the Protein Data Bank and AlphaFold model repositories. Literature retrieval is supported through web search capabilities that enable the system to cite prior engineering studies, known stabilizing mutations, and relevant mechanistic insights. Computational utilities include basic sequence analysis tools for alignment, motif scanning, and conservation scoring, as well as structural analysis functions that calculate distances to active sites, identify pocket neighborhoods, and assess solvent accessibility. A library feasibility calculator estimates variant counts based on the number and type of targeted positions. Users provide FASTA-formatted sequences, optional PDB or model files, lists of known important residues when available, experimental constraint specifications, and CSV-formatted results from completed screening rounds to enable iterative design.

User-Facing Front-End Interface

The Milestone I deliverable will be a lightweight web interface featuring a campaign setup form where users specify their objective, constraints, and upload relevant files. The dashboard output displays ranked mutation zones and candidate positions, library construction options with projected sizes and screening burden, explicit assumptions with confidence flags, and citations to supporting evidence. An iteration panel allows users to upload experimental results and receive round-two recommendations. Export functionality provides structured tables in CSV and JSON formats along with a markdown-formatted report suitable for laboratory notebooks and grant applications.

Evaluation Plan and Feasibility

The evaluation strategy combines behavioral assessment of agent quality with utility measurement of user impact. Behavioral evaluation includes consistency checks to verify that plans respect throughput constraints and library sizes are realistic, evidence grounding assessment to ensure key claims are cited or clearly labeled as assumptions, and critic effectiveness measurement by counting how often the critic identifies issues that materially improve the final plan. Utility evaluation compares time-to-plan between manual and agent-assisted workflows, collects user ratings on clarity, feasibility, confidence, and completeness using a structured rubric, and optionally benchmarks the system against known stabilizing mutations from literature to assess whether top-ranked regions overlap empirically validated hotspots. The pilot study will involve three to five users with biological backgrounds completing one to two design campaigns each using standardized prompts and evaluation criteria.

Key risks include potential hallucination of incorrect biological facts, which will be mitigated through mandatory citations, explicit assumption logging, and critic-agent validation. The system will avoid overly specific wet-lab protocols and instead focus on design-level guidance. Tool failures and incomplete structural data will be handled through graceful degradation to sequence-only modes and user-upload fallbacks. Privacy and intellectual property concerns will be addressed through local-only processing options and session-limited data retention. The project timeline spans ten weeks, with Milestone I front-end completion by week three, Milestone II multi-agent workflow by week six, Milestone III evaluation by week nine, and final presentation and report in week ten. This timeline is feasible within the class duration and prioritizes core planning, library sizing, and critique functionality over advanced computational modeling features.